Solution to Exercises

Exercise 1.1

1. The number a is positive.
2. The only even prime is 2.
   (The integer 2 is the only even prime.)
3. If $x > 0$, then $g(x) \neq 0$.
4. We change sign to both sides of the equation.
5. The polynomial $x^2 + 1$ has no real roots.
   (The equation $x^2 + 1 = 0$ has no real solutions.)
6. Multiplying both sides by a negative value of x, the inequality is reversed.
7. All solutions are odd.
   (The solution set consists of odd integers.)
8. If $\sin(\pi x) = 0$, then x is an integer.
   (The condition $\sin(\pi x) = 0$ implies that x is an integer.)
9. A matrix is invertible if its determinant is non-zero.
10. This infinite sequence has fewer negative terms.

Exercise 1.2

1. There are three special cases.
3. It does not tend to infinity.
5. Therefore c^{-1} is undefined.
7. We square both sides of the equation.
9. The equation $x^2 = y^2$ represents (describes) two orthogonal lines.
11. All matrices in this set are invertible.
13. Purely imaginary means that the real part is zero.
15. Since x is zero, I can't divide by x.
17. The function f is continuous.
19. I found fewer solutions than I expected.
21. We prove Euler's theorem.
23. The asymptotes of this hyperbola are orthogonal.
25. The solution depends on s.

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27. Thus $x = a$. (We assume that a is positive.)
29. Always remember to check the sign.

Exercise 2.1

BAD: Is 39 a prime number? [*Specific and insignificant.*]
GOOD: Why is 1 not a prime number?
BAD: What is $1/2 + 1/2^2 + 1/2^3$?
GOOD: Is a fraction the same as a rational number?
BAD: What is the real part of $2 + 3\mathrm{i}$?
GOOD: What is the real part of i^{i}?

Exercise 2.2

BAD: The set of natural numbers less than 10.
GOOD: The set of the proper subsets of a finite set.

Exercise 2.3

1. $\{1 - 2n \,:\, n \in \mathbb{N}\}$
   $(\{n \in \mathbb{Z} \,:\, n < 0,\ 2 \not| n\})$.
3. $\{(n + 1)/n \,:\, n \in \mathbb{Z} \setminus \{0\}\}$.
5. $\{z \in \mathbb{C} \,:\, |z| \geqslant 1\}$.
7. Let $F(x, y) = 0$ be the equation of a circle with the stated property. The function F must be such that $F(0, 0) = 0$.
9. No such a line goes through the origin, so we represent lines as $ax + by + 1 = 0$. The tangency condition results in a relation between a and b.

Exercise 2.4

1. The set of rational points in the open unit interval.
3. The set of rationals whose denominator is a power of 2.
5. The imaginary axis in the complex plane, excluding the origin.
7. The set of integer pairs whose first component divides the second.
9. The set of points in a euclidean space whose coordinates have zero sum.

Exercise 2.5

1. The inverse of a function.
3. The value of a function at the reciprocal of the argument.
5. The composition of a function with itself.
7. The restriction of a function to the integers.
9. The image of the irrational numbers under a function.

Exercise 2.6

1. The quotient of two fractions is a fraction. Its numerator (denominator) is the product of the numerator (denominator) of the first fraction and the denominator (numerator) of the second fraction.

3. Compute the cube of the natural numbers: 1, 8, 27, etc. Stop when either (i) you obtain your integer (which is then a cube); or (ii) you obtain a larger integer, in which case your integer is not a cube.
   If your integer is 1, then it's a cube; otherwise, perform its prime factorization. If the exponent of each prime factor is divisible by 3, then your integer is a cube.
5. Transform the equation of the circle so that one side is zero, and the other side has positive quadratic coefficients. Then substitute the coordinates of the point into the equation. If you get a negative value, then the point is inside the circle; otherwise it's outside (or on the boundary).
7. Form the matrix that has your vectors as rows (or columns). Then compute the determinant of the matrix, and verify that it is non-zero.
9. Two things must happen: (i) there is a point at which the value of the two functions is the same; (ii) at this point, the product of the derivatives of the two functions is equal to -1.

Exercise 2.7

1. Compute the value of the function at each point of the domain, collecting together the resulting values. Eliminate all duplicates, so as to turn this list into a set. Your function is surjective precisely when the cardinality of this set is the same as that of the co-domain.
   Compute the value of the function at each point of the domain, collecting together the resulting values. Now verify that each element of the co-domain also belongs to your list of values. If this is true, then your function is surjective; otherwise it isn't.

Exercise 2.8

1. Let $X = A^2 \setminus B^2$ and $Y = (A \setminus B)^2$. If $A \subset B$, then, necessarily, $A^2 \subset B^2$, and hence $X = Y = \emptyset$. If A and B are disjoint, then so are A^2 and B^2, and hence $X = Y = A^2$. We will show that in all other cases Y is a proper subset of X.
   We begin with an example. Let $A = \{1, 2\}$ and $B = \{1\}$. The pair $(2, 1)$ belongs to $A^2 \setminus B^2 = \{(1, 2), (2, 1), (2, 2)\}$ but not to $(A \setminus B)^2 = \{(2, 2)\}$, as desired. Let us generalise this argument. If A is not a subset of B, then there is an element a in $A \setminus B$, and if A and B are not disjoint, then we can find $b \in A \cap B$. Now, the pair $z = (a, b)$ belongs to X (because $z \in A^2$ and $z \notin B^2$) but doesn't belong to Y (because the second component of z is not in $A \setminus B$). We have shown that Y is a proper subset of X, as desired.

Exercise 2.11 First fix some ambient set. What are the domain and co-domain of such a function?

Exercise 2.12 Consider *infinite* sequences of non-negative integers, where the kth term represents the power of the kth prime in the prime factorisation of an integer.

Exercise 2.13

1. Let Γ be such a set. We identify a segment in Γ via its mid-point z (a pair of real numbers), length r (a positive real number), and orientation θ (an angle between 0 and π). We see that

$$\Gamma \sim \Gamma^* \qquad \text{where } \Gamma^* := \mathbb{R}^2 \times \mathbb{R}^+ \times [0, \pi).$$

Consider now the subset U of Γ consisting of all segments of unit length. Using our representation, we have

$$U \sim \{(z, r, \theta) \in \Gamma^* \,:\, r = 1\} \sim \mathbb{R}^2 \times [0, \pi).$$

In the last equivalence, we have removed the idle variable r, because its value is fixed.
2. Consider the representation of a segment.
3. Consider infinite integer sequences.
4. Consider sums of the type

$$\sum_{k=n}^{\infty} d_k 10^{-k}$$

where n is an integer, and (d_k) is the sequence of decimal digits.
5. Consider the binary digits of fractions.

Exercise 3.1

1. $a_1 + a_2 + a_3$
3. $a_2 + a_1 + a_0 + a_{-1}$
5. $a_0 + a_2 + a_3 + a_4 + a_6$.

Exercise 3.2

1. (i) An identity.
   (ii) An arithmetical identity, expressing a cube as the sum of three cubes.
3. (i) A chain of inequalities.
   (ii) Upper and lower rational bounds for the square root of 2.
5. (i) An inequality.
   (ii) An algebraic inequality in two unknowns.
   (The inequality defining the first and third open quadrants in the cartesian plane.)
7. (i) An equation.
   (ii) The cartesian equation of a parabola passing through the origin.
9. (i) An identity.
   (ii) The trigonometric formula for the sine of the difference of two angles.

Exercise 3.3

1. (i) A set.
   (ii) The set of even integers that are not divisible by 4.
3. (i) A set.
   (ii) The set of rational points on the plane, with non-integer components.
   (The set of pairs of rational numbers that are not integers.)
5. Same as Exercise 2.4.11.
7. (i) A sequence.
   (ii) An infinite sequence of sets, each containing one and the same element.
9. (i) A finite sequence.
   (ii) A finite sequence, obtained by raising consecutive natural numbers to the same even power.

Exercise 3.4

1. (i) A function.
   (ii) The real function that adds 1 to its argument.
   (The function that performs the right unit translation on the real line.)
3. (i) An identity. (A functional identity.)
   (ii) The formula for the derivative of the product of two functions.
5. (i) An integral.
   (ii) The indefinite integral of a function of two variables, performed with respect to the first variable.
7. (i) An identity. (A definition.)
   (ii) The power series of the cosine.
9. (i) A finite product of functions. (An analytic expression.)
   (ii) The product of all the partial derivatives of a function of several variables.

Exercise 3.5

1. (i) A set.
   (ii) The intersection of the inverse images of the elements of a sequence of sets.

Exercise 4.1

1. $\exists x \in X, \quad x \notin Y$
3. $\exists x, y \in X, \quad x \neq y$

Exercise 4.2

1. $\exists k \in \mathbb{Z}, \quad n = k^3$
3. $\exists x \in \mathbb{Q}, \quad f(x) = 0$
5. $\forall x \in \mathbb{Z}, \quad p(x) \neq 0$
7. $\exists y \in B, \quad \forall x \in A, \quad f(x) \neq y$
9. $\exists x, y \in A, \quad f(x) \neq f(y)$

Exercise 4.3 In part (i) there is freedom in the choice of the ambient set: the smaller the set, the simpler the predicate. (But if the set becomes too small the implication disappears, see Exercise 4.10.3.)

1. (i) For all primes p, if $p > 2$, then p is odd. (TRUE)
   (ii) For all primes p, if p is even, then $p \leqslant 2$.
   (iii) For all primes p, if p is odd, then $p > 2$. (TRUE)
   (iv) There is a prime p such that $p > 2$ and p is even.
3. (i) For all positive fractions a/b, if a/b is reduced, then b/a is reduced. (TRUE)
   (For all fractions a/b, if a/b is positive and reduced, then b/a is reduced.)
   (ii) For all positive fractions a/b, if b/a is not reduced, then a/b is not reduced.
   (iii) For all positive fractions a/b, if b/a is reduced, then a/b is reduced. (TRUE)
   (iv) There is a positive fraction a/b such that a/b is reduced but b/a is not reduced.
5. See end of Sect. 4.5.
7. (i) For all odd integers n, if n is greater than 3, then one of $n, n+2, n+4$ is not prime. (TRUE)
   (For all integers n, if n is odd and greater than 3, then one of $n, n+2, n+4$ is not prime.)
   (ii) For all odd integers n, if n, $n+2$ and $n+4$ are all prime, then $n \leqslant 3$.
   (iii) For all odd integers n, if one of n, $n+2$ and $n+4$ is not prime, then $n > 3$. (FALSE)
   (iv) There is an odd integer n such that $n > 3$, and $n, n+2, n+4$ are all prime.
9. Make sure that hypothesis and conclusion are not contradictory.

Exercise 4.4 This is the definition of nephew:

x is a nephew of y := There is z such that x is a son of z and (z is a brother of y or z is a sister of y).

Thus to define nephew we must first define of brother and sister; in turn these will require other definitions.

Exercise 4.5

(a.1) The integer 5 has a proper divisor. (FALSE)
(a.3) Every natural number properly divides its square. (FALSE)
(b.1) The integer y has a proper divisor.
(c) Consider part (b).

Exercise 4.6

1. The set of prime numbers, together with the integer 1.

Exercise 4.7 Use Theorem 4.3, Sect. 4.5.

1. (FALSE). $\exists n \in \mathbb{N}, \quad 1/n \in \mathbb{N}$.
   The reciprocal of a natural number is not a natural number.
   There is a natural number whose reciprocal is also a natural number.
3. (TRUE). $\exists x, y \in \mathbb{R}, \quad xy \neq yx$.
   The product of real numbers is commutative.
   The product of real numbers is not commutative.
5. (TRUE). $\exists m, n \in \mathbb{Z}, \quad (m+n \in 1+2\mathbb{Z}) \wedge (m \notin 2\mathbb{Z} \wedge n \notin 2\mathbb{Z})$.
   If the sum of two integers is odd, then at least one summand is even.
   There are two odd integers whose sum is odd.

Exercise 4.9 Let $G = \{1, \ldots, n\}$. Think of the computation of the predicate as the evaluation of the entries of an $n \times n$ matrix, where the (i, j)-entry is the value of 'i loves j'. Then decide the order in which the entries are accessed, e.g., first by row, then by column.

1. Let $g = n$, say. If the sentence is true, then for all $k = 1, \ldots, n$ we must verify that $\mathscr{L}(k, g) = \mathrm{T}$ (George is included), so in any case we need n function evaluations. If the sentence is false, then the minimum is 1 (we find $\mathcal{L}(1, g) = \mathrm{F}$ at the first evaluation). The maximum is n (we find $\mathcal{L}(k, g) = \mathrm{T}$ for all $k \neq n$, and $\mathcal{L}(g, g) = \mathrm{F}$).

Exercise 4.10

1. Use Theorem 4.1.
3. Consider the set of which $\mathscr{P}$ is the characteristic function.

Exercise 5.1 First translate symbols into words literally; then synthesise the meaning of the literal sentence. (It may be helpful to draw the graph of a function that has the stated property, and a function that hasn't.)

1. (i) The value of f at the origin is rational.
   (ii) $\exists r \in \mathbb{Q}, f(0) = r$.
3. (i) The function f is constant.
   (ii) $\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, f(x) = y$.
5. (i) The function f vanishes at some integer.
   (ii) $\exists x \in \mathbb{Z}, f(x) = 0$.
7. (i) The function f assumes only rational values.
   (ii) $\forall y \in \mathbb{R}, f(x) \in \mathbb{Q}$.
9. (i) The integers and the natural numbers have the same image under f.
   (ii) $\forall y \in \mathbb{Z}, \quad \exists x \in \mathbb{N}, \quad f(x) = f(y)$.

Exercise 5.3

1. The zeros of f include all even integers.
3. The function f is identically zero for negative values of the argument.
5. The function is non-zero at all rationals, with the possible exception of 0.
7. The function f vanishes infinitely often on the natural numbers.

Exercise 5.4

1. $\forall x \in \mathbb{R}, \quad f(x) \neq 0.$
3. $\forall x \in \mathbb{R}, \quad f(x) = 0 \Rightarrow x = 0.$
5. $\forall x \in \mathbb{Z}, \quad f(\pi x) = 0.$
7. $\forall \varepsilon \in \mathbb{R}^+, \quad \exists x \in (-\varepsilon, \varepsilon) \smallsetminus \{0\}, \quad f(x) = 0.$

Exercise 5.5

1. (a) If $-f$ is increasing, then f is decreasing. (TRUE)
   (b) If $-f$ is not increasing, then f is not decreasing. (TRUE)
   (c) There is a decreasing function f such that $-f$ is not increasing.

3. (a) If f is monotonic, then $|f|$ is increasing.
   (FALSE, e.g., $f(x) = x$)
   (b) If f is not monotonic, then $|f|$ is not increasing.
   (FALSE, e.g., $f(x) = e^x$ for $x < 0$ and $-e^x$ for $x \geqslant 0$.)
   (c) There is a non-monotonic function whose absolute value is increasing.
5. (a) If f is surjective, then f is unbounded. (TRUE)
   (b) If f is not surjective, then f is bounded.
   (FALSE, e.g., $f(x) = \lfloor x \rfloor$.)
   (c) There is an unbounded function which is not surjective.

Exercise 5.6

1. (i) For all real functions f, if f is differentiable, then f is continuous. (TRUE)
   (ii) For all real functions f, if f is discontinuous, then f is not differentiable.
   (iii) Every continuous real function is differentiable. (FALSE)
   (iv) There is a real function f which is differentiable but not continuous.
3. (i) For all real functions f and g, if f and g are odd, then $f + g$ is odd. (TRUE)
   (ii) For all real functions f and g, if $f + g$ is not odd, then one of f or g is not odd.
   (iii) For all real functions f and g, if $f + g$ is odd, then f and g are odd. (FALSE)
   (iv) There are odd real functions f and g such that $f + g$ is not odd.

Exercise 5.8

1. A smooth bounded odd function, which vanishes infinitely often and approaches zero alternating sign as the argument tends to infinity.
3. This is a step function defined only for positive arguments. The steps have unit length, while their height increases monotonically.
5. A differentiable even function, with a global minimum at the origin and no maximum. For positive arguments the function increases monotonically, approaching a positive limit.

Exercise 5.10

1. The logarithm.
3. The identity.
5. The sine of the square of the argument.
7. The square of the arc-tangent.

Exercise 5.11

1. $\exists k \in \mathbb{N},\quad a_k \neq b_k$
3. $\forall n \in \mathbb{N},\quad \exists k \in \mathbb{N},\quad a_k \neq a_{k+n}$
5. $\forall k \in \mathbb{N},\quad \exists j \in \mathbb{N},\quad a_{k+j} < 0$
7. $\forall \varepsilon \in \mathbb{R}^+,\quad \exists k \in \mathbb{N},\quad |a_k| < \varepsilon$
9. $\forall n \in \mathbb{N},\quad \exists k \in \mathbb{Z} \smallsetminus \{0\},\quad a_{n+k} = a_k.$

Exercise 6.2

1. When we say that the price of petrol, say, on Thursday, is 10 % higher that the price on Wednesday, we mean that the Thursday price is equal to the Wednesday price plus one tenth of the Wednesday price, so it's $(1 + 1/10) = 11/10 = 1.1$ times the Wednesday price. Likewise, if we say that the Friday price is 10 % lower than the Thursday price, that means it's the Thursday price minus one tenth of the Thursday price (*not* one tenth of the Wednesday price!); so the Friday price is $(1 - 1/10) = 9/10 = 0.9$ times the Thursday price. In all, the Friday price is the Wednesday price times 11/10 times 9/10, which is 99/100 = 0.99 times the Wednesday price—nearly the Wednesday price but not quite.
2. Let $\mathbb{R}_{\geqslant 0}$ be the set of non-negative real numbers, representing the values of an observable (e.g., the price of a commodity). We consider the function

$$f_p : \mathbb{R}_{\geqslant 0} \to \mathbb{R}_{\geqslant 0}$$

where $f_p(x)$ is the result of varying x by a given percentage amount p, followed by a variation by $-p$. The parameter p is a real number, which we require to be in the range $-100 \leqslant p \leqslant 100$, to ensure that the process does not result in negative quantities, i.e., that $f_p(x) \in \mathbb{R}_{\geqslant 0}$.
A single change of x by p % corresponds to the mapping

$$x \mapsto x\left(1 + \frac{p}{100}\right).$$

If this transformation is followed by a percentage change with the opposite sign, we obtain

$$f_p(x) = x\left(1 + \frac{p}{100}\right)\left(1 - \frac{p}{100}\right) = x\left[1 - \left(\frac{p}{100}\right)^2\right].$$

For fixed p, this is a linear function of x, whose coefficient depends on $|p|$. If we exclude the trivial case $p = 0$, we have $f_p(x) < x$, which means that at the end of the process the value of x has become smaller. The decrease is maximal if $p = \pm 100$, in which case the final value is zero, irrespective of the initial value.

Exercise 6.5 Imagine that there are 100 boxes, instead of three; the presenter opens 98 boxes, all of them empty.

Exercise 6.7

1. You should identify at least three definitions.
2. Construct a sequence of intervals with given mid-point and length converging to zero.
3. The domain of a 'random variable' is the space of elementary outcomes. For each outcome, the 'random variable' gives us a real number.
4. Begin with the following lineup: $n!$, n^{10}, 2^n. Consider $n^{\log(n)}$ versus $\log(n)^n$.

Exercise 7.1

1. A point on the first line is represented by the position vector

$$\mathbf{l}(\lambda) = (4, 5, 1)^T + \lambda\,(1, 1, 1)^T = (4 + \lambda, 5 + \lambda, 1 + \lambda)^T$$

for some real number λ. Similarly, a point on the second line has position vector

$$\mathbf{m}(\mu) = (5, -4, 0)^T + \mu\,(2, -3, 1)^T = (5 + 2\mu, -4 - 3\mu, \mu)^T$$

for some real μ. For the lines to intersect, there must exist values of λ and μ for which the two position vectors are the same, namely

$$\mathbf{l}(\lambda) = \mathbf{m}(\mu). \tag{10.3}$$

The above vector equation corresponds to three scalar equations:

$$4 + \lambda = 5 + 2\mu \tag{10.4}$$

$$5 + \lambda = -4 - 3\mu \tag{10.5}$$

$$1 + \lambda = \mu. \tag{10.6}$$

Eliminating μ from Eqs. (10.4) and (10.6), we obtain

$$4 + \lambda = 5 + 2(1 + \lambda)$$

which yields the solution $\lambda = -3$, $\mu = -2$. This solution also satisfies Eq. (10.5), and hence it satisfies all three equations. Substituting these values in Eq. (10.3) we obtain

$$\mathbf{l}(-3) = \mathbf{m}(-2) = (1, 2, -2)^T$$

which is the position vector of the common point of the two lines. □

3. Assign symbols to the line and the parabola. Use the same symbols in the corresponding equations.

Exercise 7.2

(a) The statement of the theorem is imprecise in several respects.

The nature of the numbers x and y is not specified (the inequality would be meaningless for complex numbers).
The case in which one of x or y is zero should be excluded, since in this case the left-hand side of the inequality is undefined.
The statement is false unless the inequality is made non-strict; indeed the equality holds for infinitely many values of x and y.

There are several flaws in the proof.

The basic deduction is carried out in the wrong direction, which proves nothing. (Proving that $P \Rightarrow$ TRUE gives no information about P.)
The assertion 'the last equation is trivially true' is, in fact, false for $x = y$.
The writing is inadequate, without sufficient explanations, and also imprecise (the expression $(x-y)^2 > 0$ is an inequality, not an equation).

(b) Revised statement:

THEOREM: *For all nonzero real numbers x and y, the following holds:*

$$\frac{x^2+y^2}{|xy|} \geqslant 2.$$

PROOF: Let x and y be real numbers, with $xy \neq 0$. We shall deduce our result from the inequality $(x \pm y)^2 \geqslant 0$. We begin with the chain of implications:

$$(x \pm y)^2 \geqslant 0 \quad \Rightarrow \quad x^2 \pm 2xy + y^2 \geqslant 0 \quad \Rightarrow \quad x^2 + y^2 \geqslant \mp 2xy.$$

Now, since xy is non-zero, we divide both sides of the rightmost inequality by $|xy|$, to obtain

$$\frac{x^2+y^2}{|xy|} \geqslant \mp 2\frac{xy}{|xy|}.$$

The expression $xy/|xy|$ is equal to 1 or -1, and by choosing an appropriate sign we can ensure that $\pm xy/|xy| = 1$. Our proof is complete. □

Exercise 7.3

2. The flaw is subtle: it's related to the incorrect inversion of the logarithmic function. In Sect. 7.7.3 we dealt with the same problem in a simpler setting.

Exercise 7.4

1. Let X be a compact set, and let f be a continuous function over X. RTP: f is uniformly continuous.
3. Let $f(x)$ be a polynomial with integer coefficients, and assume that f is not a constant. RTP: There is an integer n such that $f(n)$ is not prime.
5. Let D be the interior of the unit disc, and let f be a bi-unique analytic mapping of D into itself. RTP: f is a linear fractional transformation.
7. Let P be a Pisot substitution, let M be its incidence matrix, and let f be the characteristic polynomial of M. RTP: f is irreducible over $\mathbb{Q}$.
9. Let D be a differential of the second or third kind. RTP: There is a normal differential N such that $D - N$ is a differential of the first kind.

Exercise 7.5 Consider Euclid's algorithm.

Exercise 8.1 The difficulty in a computer-assisted proof consists in converting the two sides of the identities (8.4) to the same form. In the following Maple codes the function `expand` is used for this purpose. In each case, the output of the last expression is the boolean constant TRUE.

1.
```
a:=k->2*k-1:
F:=n->n^2:
a(1)=F(1) and F(k)+a(k+1)=expand(F(k+1));
```

3. The function `sum` performs symbolic summation.

```
a:=k->k^3:
F:=n->(sum(j,j=1..n))^2:
a(1)=F(1) and expand(F(k)+a(k+1)-F(k+1))=0;
```

5. The base case is $n = 2$.

Exercise 8.4

1. The formula is given in Sect. 2.3.1.
3. Define $f^n = \underbrace{f \circ f \circ \cdots \circ f}_{n}$, and $x_n = f^n(x)$, with $x_0 = x$. Then

$$(f^n(x))' = \prod_{k=0}^{n-1} f'(x_k).$$

Exercise 8.5

1. The base case is not proved.

Exercise 9.1

1. Divide the square into four suitable regions.
2. Consider the function giving the number of friends of a person.
3. The boxes are the elements of the set; what are the objects?

Exercise 9.2

1. The denominator of $f(x)$ vanishes at $x = (-1 \pm \sqrt{5})/2$. Hence the domain of the function should be restricted to $\mathbb{R} \smallsetminus \{(-1 \pm \sqrt{5})/2\}$.
3. The value of the function at a/b is not well-defined, because a and b are not necessarily co-prime. Such a value should be changed to $|a - b|/\gcd(a, b)$. Alternatively, one could change the domain to the set of reduced rationals.

Exercise 9.3

1. FAULTS:
   The integer m must be positive, otherwise the summation is undefined. Alternatively, the sum should be defined to be 0 if $m < 1$.
   The set B is undefined. In particular, we cannot conclude that the reciprocal of each term of the sequence is well-defined.

The co-domain of f is not necessarily B, even if the sum is well-defined.
The symbol a is introduced but not used.
To improve clarity, the dependence of f on the sequence should be made explicit, as a parameter or a second argument. For coherence, this quantity should be called b, not a. Stating that the sequence is infinite would also be helpful.

REVISED STATEMENT:
Given any infinite sequence $b = (b_1, b_2, \ldots)$ of non-zero complex numbers, we define the function f_b as follows:

$$f_b : \mathbb{N} \to \mathbb{C} \qquad m \mapsto \sum_{k=1}^{m} b_k^{-1}.$$

(The set $\mathbb{C}$ could be replaced by any field.)

Exercise 9.4

1. Consider the number of twin primes smaller than a given bound.
2. Is the representation of an even integer as a sum of two primes unique?
3. Consider the smallest non-negative integer t for which the sequence with initial condition $x_0 = n$ has $x_t = 1$.

Exercise 10.1

1. The writing is clumsy; remove reference to project and author in the first sentence; the acronym CF is not used; the verb 'examine' is over-used.
2. Bad first sentence: 'an example of this is that'; the numeral '2' is inappropriate; the main topic (accelerator modes) should be mentioned in the opening sentence; a symbol is introduced but not used; the last two sentences could be merged into one.
3. Inappropriate and heavy use of symbols obscures the main message; introduce a concise symbol for $GQ(s, s)$; the conjecture is formulated twice; the last sentence is clumsy.
4. The first sentence should be re-arranged; some symbols are unnecessary, e.g., those specifying the range of k-values.

Exercise 10.3 How does k depend on n when n is large? Consider Stirling's formula[1] for the factorial.

Exercise 10.4

1. The following Maple code generates the sequence (a_n):

```
N:=10000:
a:=array(1..N,[1]):
for n from 2 to N do
```

[1] James Stirling (Scottish: 1692–1770).

```
    if isprime(n^2+n+41) then
      a[n]:=a[n-1]+1
    else
      a[n]:=a[n-1]
    end if
end do:
```

We start with the lower bound $f(n) = \sqrt{n}$, while the counterexample (7.8) shows that $g(n) \leqslant 40n/41$ for $n \geqslant 41$. The following code displays the ratios $f(n)/a_n$ and $g(n)/a_n$ within the same plot:

```
with(plots):
f,g:=n->sqrt(n),n->n*40/41:
display(plot([seq([n,f(n)/a[n]],n=1..N)]),
         plot([seq([n,g(n)/a[n]],n=1..N)],color=blue));
```

Sharpen these bounds by experimenting with fractional powers and logarithms. (In particular, choose g so that $g(n)/n \to 0$.) A pertinent reference is Golbach's theorem (1752), which states that no non-constant polynomial with integer coefficient can be prime for all (or all sufficiently large) values of the indeterminate, see [15, Theorem 21].

2. This code generates and plots the sequence (z_n):

```
a:=n->n^2/50:
N:=5000:
TwoPiI:=evalf(2*Pi*I):
z:=array(1..N):
z[1]:=evalf(exp(TwoPiI*a(1))):
for n from 2 to N do
  z[n]:=z[n-1]+evalf(exp(TwoPiI*a(n))):
end do:
plot([seq([Re(z[n]),Im(z[n])],n=1..N)]):
```

To increase speed, the exponentials are converted to floating-point with `evalf`. These sequences produce beautiful patterns, but the latter will become apparent only if several thousand terms are plotted.
Gauss first studied these sums in the case $a_n = n^2/p$, $N = p$, with p prime. This is a **quadratic Gauss' sum**, which can be the starting point for a bibliographical search.

3. (a) If $a_n \in \mathbb{N} \setminus A$, then q_n is divisible by 2, 3, or 5.
(b-c) The following code generates the sequence (a_n):

```
N:=500:
a:=[seq(nextprime(10^(n-1))-10^(n-1),n=1..N)]:
```

This computation is very time-consuming, as it deals with huge integers.
Define the sequences of sets

$$A_n = \{a_k \,:\, k \leqslant n\}, \qquad W_n = A \setminus A_n, \qquad n \geqslant 1,$$

where A is given in (10.2). Let a_n^+ be the largest element of A_n and let a_n^- be the smallest element of W_n. Both sequences are non-decreasing. The set $f(\mathbb{N})$ is unbounded if and only if $a_n^+ \to \infty$, whereas the condition $a_n^- \to \infty$ is seen to be equivalent to $f(\mathbb{N}) = A$.
The following code generates the sequences $(a_n^\pm)$:

```
A:={$1..N} minus
    select(n->igcd(n,10)>1 or (n mod 3)=2, {$1..N}):
ap,am:=array(1..N),array(1..N):
for n to N do
   {op(a[1..n])}:
   A minus %:
   ap[n]:=max(%%):
   am[n]:=min(%%)
end do:
```

You may consider replacing a_n^+ with the less volatile arithmetical mean of the elements of A_n, generated with the command `add(k,k=a[1..n])/n`. A plot of few hundred terms of this sequence should convince you that $f(\mathbb{N})$ is unbounded, but you must decide if your data provide enough support to the conjecture that $f(\mathbb{N}) = A$.
The bibliography could refer to expository works on the randomness of prime numbers, e.g., [38, Chap. 3].

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